Calculating the Reaction Forces on the Ankle, Knee, and Hip during the Stance Period of Gait

Performed and Written by Ani Gregori Asadourian and Anne Jumper

April 20, 2025

Introduction

The bipedal gait cycle is the series of movements during walking from which one foot makes contact with the ground to when that same foot contacts the ground again. It consists of the stance phase, when the foot being analyzed is in contact with the ground, and the swing phase when that foot is not in contact with the ground. During this process, one propels the center of gravity in the direction of motion, creating net reaction forces and moments on the leg's joints— the ankle, knee, and hip. By using motion capture, a force plate, and anthropometry to collect data on the kinematics of the leg's motion during gait, developing a link-segment model of the leg, and using the 2D equations of motion (kinetics) to analyze each segment, we can determine these forces and moments in the joints both at an instant during gait and during the entire stance phase of gait. This method of using kinematics of motion to solve kinetic equations is called inverse dynamics analysis. These results are significant in reducing injury risk and improving assistive devices, allowing for better performance both athletically and in everyday life.

Methods

In order to solve the equations of motion using inverse dynamics analysis, data on the kinematics— velocity and acceleration— of the leg during the gait cycle is required. This data is collected by analyzing the gait of a participant in the positive x-direction using motion capture, which requires the placement of 13 reflective markers on the shoulders, hips, knees, ankles, big toes, heels, and back of a participant. Their gait is then recorded using VICON cameras, which track the motion of the reflective markers in the x- and z-directions to calculate the linear (x and z) and angular accelerations of each segment— the thigh, shank, and foot. A force plate is also used to determine the ground reaction force (GRF) during stance and the center of pressure (COP) under the foot (average point of application of the GRF). The starting point of the participant's walk ensures that their right foot lands directly on the force plate during stance to record an accurate GRF and COP. Existing data gathered via anthropometry, the study of the measurements of the human body, is used to develop an accurate link-segment model of the leg and to determine the center of mass (COM) location, mass, and moment of inertia of each segment.

The data collected is then used to solve the equations of motion in the xz-plane for the reaction forces and moments at each joint: the ankle, knee, and hip. They are: $\Sigma F_x = m(a_G)_x$, $\Sigma F_z = m(a_G)_z$, and $\Sigma M_G = I_G\alpha$ and are applied to each of the leg's linkages. The motion capture data provides the kinematic information about each of the linkages over the stance period, including the linear accelerations in the x and z directions of the COM and angular accelerations about the COM. The motion capture also details the x and z positions of the joints and the force plate reveals the GRF acting on the foot and the foot's COP. Reading this data into MatLab, the system of kinetic equations can then be solved to find the reaction forces acting on each joint. For example, after reading in the above data for the foot, we set up the three kinetic equations to solve using inverse dynamics. First is the reaction force in the x direction for the ankle using $\Sigma F_x = m(a_G)_x$, which we find by multiplying the foot's mass by the linear COM acceleration in the x-direction and subtracting the contribution of the GRF to the net force. This same calculation is performed in the z-direction using $\Sigma F_z = m(a_G)_z$, now accounting for both the linear

acceleration in the z direction from motion capture and the acceleration due to gravity, and subtracting the GRF in the z-direction. Finally, the moment about the ankle is found using $\Sigma M_G = I_G \alpha$. The moment of inertia about the foot's COM is multiplied by the angular acceleration. To this, we add the torque contributions from the reaction forces: the x-direction reaction force multiplied by its vertical lever arm (the vertical distance from the foot's COM to the ankle), and the z-direction reaction force multiplied by its horizontal lever arm (the horizontal distance from foot's COM to the ankle). We then subtract the torques caused by the GRFs: the z-component of the GRF multiplied by its lever arm (the horizontal distance from the foot's COP to the foot's COM) and the x-component of the GRF multiplied by its lever arm (the z-position of the foot's COM, since the z-position of the COP is zero). The calculation of these three kinetic equations is repeated for the knee and hip, now excluding the contribution of the GRFs, which do not act directly on the knee and hip, but including the contributions of the reaction forces for the attached linkages below. That is, the calculations for the knee include the reaction forces and moments for the ankle, since the ankle is attached to the shank, and the calculations for the hip include the reaction forces and moments for the knee, since the knee is attached to the thigh. Using MatLab to read the motion capture data into arrays, these reaction forces and moments are calculated both at a single moment during gait and over the full stance period of gait and plotted to see how their values change throughout stance.

Results

Figure 1

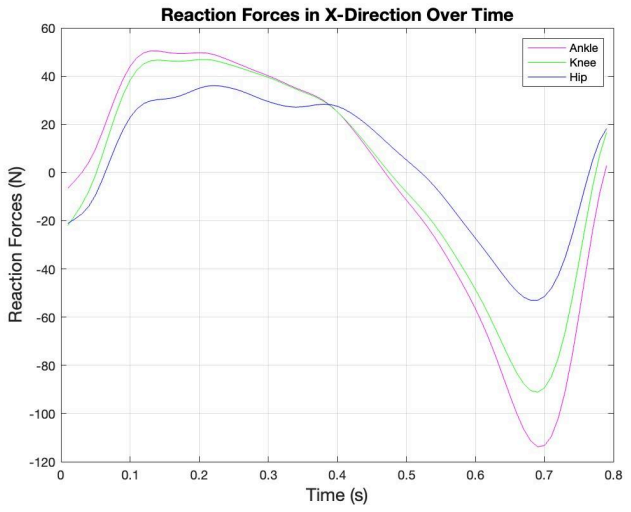


Figure 2

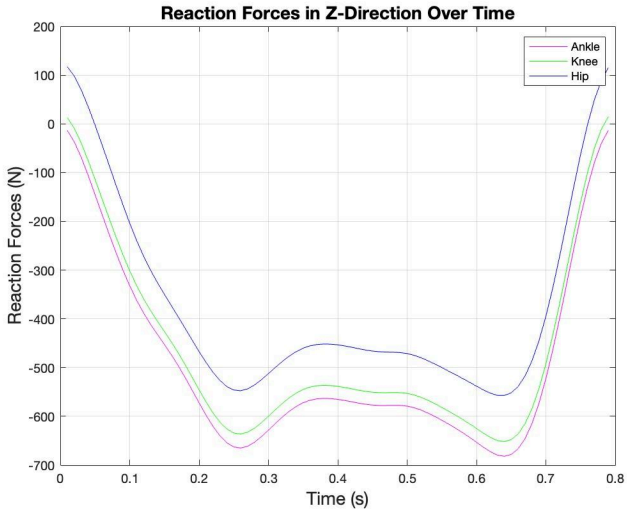


Figure 3

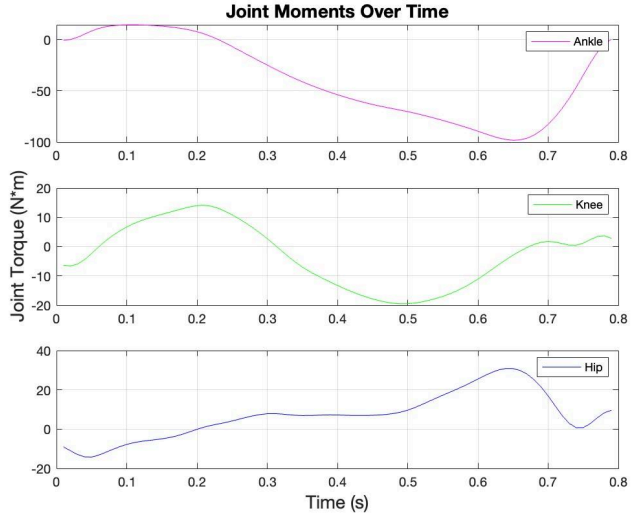


Figure 1 shows the reaction forces (measured in Newtons) acting on the ankle, knee, and hip in the x-direction (horizontal) and how they change throughout the stance phase of gait (time in seconds). Figure 2 shows the reaction forces (measured in Newtons) acting on the ankle, knee, and hip in the z-direction (vertical) and how they change throughout the stance phase of gait (time in seconds). Figure 3 shows the reaction torques or moments (measured in Newton meters) acting on the ankle, knee, and hip in three separate plots and how they change throughout the stance phase of gait (time in seconds).

Discussion

Analysis of the reaction forces at the joints reveals that the ankle experiences the highest magnitude reaction forces in the x- and z-directions and the largest reaction torque. This is a reasonable fact since the ankle supports the weight of the rest of the body, adding to the force and torque experienced at this joint. Additionally, at around 0.65 seconds, there is a large negative peak in the reaction torque at the ankle. This is during the end of the stance phase when the toe is pushing off the ground, so all the reaction force from the ground is concentrated at the toe, well in front of the ankle, tending to cause dorsiflexion (pointing of the toes up, positive moment). To oppose this external dorsiflexion moment, the calf muscles contract to generate an internal plantarflexion moment (pointing of the toes down, negative). So, the calf muscle contraction and concentration of the GRF at the toe (long lever arm from the ankle) contribute to the high magnitude of the reaction torques and forces at the ankle during terminal stance.

Unlike the ankle moment, the knee plot of the knee's reaction torque has two extrema. The first is positive and represents the external flexion (i.e. bending) moment on the knee which occurs as body weight is loaded onto the flat foot just after the heel strikes the ground. During this phase, the GRF passes behind the knee, tending to flex it. Thus, the quadriceps are engaged to counteract flexion, generating an internal extension moment (positive) to counteract this flexion moment (negative). The second is negative and occurs as the body propels forward off of the ground from the toe. At this time, the GRF's application moves towards the toe in front of the knee, tending to extend the knee. The hamstrings and gastrocnemius (calf muscle) are activated as a result to counter the external extension moment (positive) by creating an internal flexion moment (negative).

The hip experiences a large positive torque during terminal stance. At this instant, the heel is lifted and one propels their body forward, pushing off from the ground on their toe. As the torso moves forward ahead of the leg, the hip is extending and the GRF passes well behind the hip. This creates an external extension moment at the hip (negative), forcing the body to generate an internal flexion moment (positive) to resist additional hip extension.

Comparing this hip moment to that of the ankle, the peak ankle moment is larger than that of the hip as the ankle is more efficient in propelling the body forward by pushing off from the ground due to its ability to store elastic energy, better leverage, proximity to the ground, and due to the smaller mass it needs to move. However, calf strength is reduced with age, causing older individuals to compensate by using their hips more, resulting in an increased hip moment peak.

